Topology II

Lecture notes, Chapters 9–14 (Algebraic Topology)

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Chapter 9. The Fundamental Group

Algebraic topology attaches algebraic objects to spaces so that topological questions (are two spaces homeomorphic? does a map have a fixed point?) become algebraic ones. The first such object is the fundamental group. Throughout, $I = [0, 1]$.

9.1 Homotopy of paths

Maps $f, f' : X \to Y$ are **homotopic**, $f \simeq f'$, if there is a continuous $F : X \times I \to Y$ with $F(x, 0) = f(x)$, $F(x, 1) = f'(x)$. Paths $f, f' : I \to X$ from x_0 to x_1 are **path homotopic**, $f \simeq_p f'$, if moreover $F(0, t) = x_0$ and $F(1, t) = x_1$ for all t. Both are equivalence relations; $[f]$ denotes the path-homotopy class.

Example 9.1

In a convex subset $A \subset \mathbb{R}^n$ any two paths with the same endpoints are path homotopic by the straight-line homotopy $F(s, t) = (1 - t)f(s) + tg(s)$.

If f ends where g begins, the **product** $f * g$ runs through f on $[0, \frac{1}{2}]$ and g on $[\frac{1}{2}, 1]$ (pasting lemma). It is well defined on classes: $[f] * [g] = [f * g]$.

Theorem 9.1 (Groupoid properties)

The operation $*$ on path-homotopy classes is associative (when defined), has identities $[e_x]$ (constant paths), and inverses $[\bar{f}]$, $\bar{f}(s) = f(1 - s)$.

Proof. The key observation: if $k : I \to I$ is continuous with $k(0) = 0$, $k(1) = 1$, then $f \circ k \simeq_p f$ (straight-line homotopy in I). Associativity: $(f * g) * h$ is a reparametrization of $f * (g * h)$. Identities similarly. For inverses, $F(s, t) = f(2s(1 - t))$ for $s \le \frac{1}{2}$ and $f((2 - 2s)(1 - t))$ for $s \ge \frac{1}{2}$ contracts $f * \bar{f}$ to e_{x_0}. ∎

9.2 The fundamental group

Definition (Fundamental group)

For $x_0 \in X$, $\pi_1(X, x_0)$ is the set of path-homotopy classes of loops at x_0 with the operation $*$.

A path α from x_0 to x_1 induces an isomorphism $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$, $[f] \mapsto [\bar{\alpha}] * [f] * [\alpha]$. So for path connected X the group does not depend on the base point up to isomorphism (but the isomorphism depends on α unless π_1 is abelian). X is **simply connected** if path connected with trivial π_1; then any two paths with the same endpoints are path homotopic. Convex subsets of $\mathbb{R}^n$ are simply connected.

A continuous $h : (X, x_0) \to (Y, y_0)$ induces $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $[f] \mapsto [h \circ f]$, with $(k \circ h)_* = k_* \circ h_*$ and $(\mathrm{id})_* = \mathrm{id}$. Hence homeomorphisms induce isomorphisms: π_1 is a topological invariant.

9.3 Covering spaces

Definition (Covering map)

A continuous surjection $p : E \to B$ is a **covering map** if every $b \in B$ has an open neighbourhood U that is **evenly covered**: $p^{-1}(U)$ is a disjoint union of open sets V_α each mapped homeomorphically onto U by p.

Examples: $p : \mathbb{R} \to S^1$, $p(x) = (\cos 2\pi x, \sin 2\pi x)$; $z \mapsto z^n$ on S^1; products of covering maps such as $\mathbb{R}^2 \to S^1 \times S^1$. A covering map is a local homeomorphism and an open map; the converse fails ($\mathbb{R}_+ \to S^1$).

A **lifting** of $f : X \to B$ is $\tilde{f} : X \to E$ with $p \circ \tilde{f} = f$.

Lemma 9.2 (Path and homotopy lifting)

Let $p : (E, e_0) \to (B, b_0)$ be a covering. (a) Every path in B starting at b_0 has a unique lifting starting at e_0. (b) Every map $F : I \times I \to B$ with $F(0,0) = b_0$ has a unique lifting $\tilde{F}$ with $\tilde{F}(0,0) = e_0$; if F is a path homotopy, so is $\tilde{F}$.

Proof. (a) By the Lebesgue number lemma subdivide I so that each piece maps into an evenly covered set, and lift piece by piece using the local inverses of p. Uniqueness: two liftings agree on a set that is open and closed in I. (b) Do the same on a grid of small squares in lexicographic order; the edges $\{0\} \times I$ and $\{1\} \times I$ map to points, and their liftings are connected subsets of the discrete fibre, hence constant. ∎

Theorem 9.3 (Lifting correspondence)

Define $\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$ by $\phi([f]) = \tilde{f}(1)$, the endpoint of the lift starting at e_0. It is well defined; if E is path connected it is surjective, and if E is simply connected it is bijective.

Proof. Well defined by homotopy lifting. Surjective: join e_0 to e_1 in E and push down. Injective: if $\tilde{f}(1) = \tilde{g}(1)$, then $\tilde{f} \simeq_p \tilde{g}$ in the simply connected E, and composing with p gives $f \simeq_p g$. ∎

9.4 The fundamental group of the circle

Theorem 9.4

$\pi_1(S^1, b_0) \cong \mathbb{Z}$.

Proof. Use $p : \mathbb{R} \to S^1$ with $e_0 = 0$, so $p^{-1}(b_0) = \mathbb{Z}$ and $\phi : \pi_1 \to \mathbb{Z}$ is a bijection ($\mathbb{R}$ is simply connected). It is a homomorphism: if $\tilde{f}$ ends at n and $\tilde{g}$ at m, then $\tilde{f} * (n + \tilde{g})$ lifts $f * g$ and ends at $n + m$. ∎

More generally, for a covering $p : (E, e_0) \to (B, b_0)$, p_* is injective, the lifting correspondence induces a bijection from the right cosets of $H = p_*\pi_1(E, e_0)$ onto $p^{-1}(b_0)$, and a loop f lifts to a loop iff $[f] \in H$.

9.5 Retractions and fixed points

$A \subset X$ is a **retract** if there is a continuous $r : X \to A$ with $r|_A = \mathrm{id}$. Then $i_* : \pi_1(A) \to \pi_1(X)$ is injective (since $r_* i_* = \mathrm{id}$).

Theorem 9.5 (No-retraction theorem)

S^1 is not a retract of the closed disc B^2.

Proof. $\pi_1(S^1) = \mathbb{Z}$ cannot inject into $\pi_1(B^2) = 0$. ■

A loop $h : S^1 \to X$ is **nulhomotopic** iff it extends to B^2 iff h_* is trivial.

Theorem 9.6 (Brouwer fixed point theorem, $n = 2$)

Every continuous $f : B^2 \to B^2$ has a fixed point.

Proof. If $f(x) \neq x$ for all x, the ray from $f(x)$ through x meets S^1 in a point $r(x)$, and r is a retraction of B^2 onto S^1. ■

Corollary 9.7

A 3×3 matrix with positive entries has a positive eigenvalue (apply Brouwer to $x \mapsto Ax/\|Ax\|_1$ on the triangle $\{x \geq 0, \sum x_i = 1\}$). A nonvanishing vector field on B^2 points directly outward and directly inward at some boundary points.

9.6 The fundamental theorem of algebra and Borsuk–Ulam

Theorem 9.8 (Fundamental theorem of algebra)

A nonconstant complex polynomial has a root.

Proof. For monic $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ with $\sum|a_i| < 1$ (rescale z), the map $z \mapsto z^n$ on S^1 induces multiplication by n on π_1, hence is not nulhomotopic, and the homotopy $z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$ never vanishes on S^1. If p had no root in B^2, $p|_{S^1}$ would extend to $B^2 \to \mathbb{R}^2 - 0$, hence be nulhomotopic in $\mathbb{R}^2 - 0$, a contradiction. ■

Theorem 9.9 (Borsuk–Ulam, $n = 2$)

For every continuous $f : S^2 \to \mathbb{R}^2$ there is x with $f(x) = f(-x)$. Equivalently there is no antipode-preserving map $S^2 \to S^1$.

Proof. An antipode-preserving $h : S^2 \to S^1$ restricted to the equator would be a nulhomotopic (it extends over a hemisphere) antipode-preserving loop; but an antipode-preserving map $S^1 \to S^1$ induces an odd, hence nonzero, multiple on π_1 (lift and use $h(-x) = -h(x)$ to see that the lift of half the circle ends at a half-odd-integer). ■

Consequence (bisection theorem): two bounded polygonal regions in the plane can be simultaneously bisected by a single line. Also, no subset of $\mathbb{R}^2$ is homeomorphic to S^2.

9.7 Deformation retracts and homotopy type

Lemma 9.10

If $h, k : X \to Y$ are homotopic via H with $\alpha(t) = H(x_0, t)$, then $k_* = \hat{\alpha} \circ h_*$. In particular homotopic maps fixing the base point throughout induce the same homomorphism.

$A \subset X$ is a **deformation retract** if the identity of X is homotopic to a retraction onto A by a homotopy fixing A pointwise; then the inclusion induces an isomorphism on π_1. Examples: S^{n-1} is a deformation retract of $\mathbb{R}^n - 0$; the theta space is a deformation retract of the plane minus two points. Maps $f : X \to Y$, $g : Y \to X$ with $g \circ f \simeq \mathrm{id}_X$, $f \circ g \simeq \mathrm{id}_Y$ make X, Y **homotopy equivalent**, and then f_* is an isomorphism (by the lemma). A space homotopy equivalent to a point is **contractible**.

9.8 The fundamental groups of spheres and some surfaces

Theorem 9.11

If $X = U \cup V$ with U, V open, $U \cap V$ path connected and $x_0 \in U \cap V$, then $\pi_1(X, x_0)$ is generated by the images of $\pi_1(U, x_0)$ and $\pi_1(V, x_0)$. Hence S^n is simply connected for $n \geq 2$.

Proof. Subdivide a loop so each piece lies in U or in V, and connect the subdivision points to x_0 by paths in $U \cap V$; the loop becomes a product of loops each lying in U or in V. For S^n take U, V the sphere minus the north or south pole, each homeomorphic to $\mathbb{R}^n$, with connected intersection when $n \geq 2$. ■

Example 9.2

(a) $\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$; hence $\pi_1(T^2) \cong \mathbb{Z} \times \mathbb{Z}$.
(b) The projective plane $P^2 = S^2/(x \sim -x)$ has the 2-fold covering $S^2 \to P^2$ with simply connected total space, so $\pi_1(P^2) \cong \mathbb{Z}/2$.
(c) $\mathbb{R}^2$ and $\mathbb{R}^n$ $(n \neq 2)$ are not homeomorphic ($\mathbb{R}^n$ minus a point is simply connected for $n \geq 3$); S^2, T^2, P^2 are pairwise non-homeomorphic. The figure eight has non-abelian fundamental group (Chapter 11).

Chapter 10. Separation Theorems in the Plane

We apply the fundamental group to the topology of the plane. It is convenient to work in $S^2 = \mathbb{R}^2 \cup \{\infty\}$, where $S^2 - \{\text{point}\} \cong \mathbb{R}^2$. An **arc** is a space homeomorphic to $[0,1]$ and a **simple closed curve** one homeomorphic to S^1. A set C **separates** X if $X - C$ is not connected.

10.1 The Jordan separation theorem

Lemma 10.1 (Nulhomotopy lemma)

Let $a, b \in S^2$ and A a compact space. If $f : A \to S^2 - \{a, b\}$ is continuous and a, b lie in the same component of $S^2 - f(A)$, then f is nulhomotopic as a map into $S^2 - \{a, b\}$.

Idea. Put a at ∞; then $S^2 - \{a, b\} \cong \mathbb{R}^2 - \{0\}$, which deformation retracts to S^1. Since $f(A)$ is compact and b can be joined to ∞ avoiding $f(A)$, the map $x \mapsto (f(x) - b)/\|f(x) - b\|$ is homotopic to a map into a small arc, by moving b along a path to a point far away. ■

The basic tool is the following consequence of covering space theory.

Theorem 10.2 (Two-component lemma)

Let $X = U \cup V$ with U, V open and $U \cap V = A \cup B$, where A, B are disjoint, open and path connected. Let α be a path in U and β a path in V, both from $a \in A$ to $b \in B$. Then the loop $\alpha * \bar{\beta}$ generates an infinite cyclic subgroup of $\pi_1(X, a)$; in particular $\pi_1(X) \neq 0$.

Idea. Build a covering space E of X by gluing countably many copies $U \times \{2n\}$ and $V \times \{2n+1\}$, identifying A in $U \times \{2n\}$ with A in $V \times \{2n-1\}$ and B in $U \times \{2n\}$ with B in $V \times \{2n+1\}$. The loop $(\alpha * \bar{\beta})^m$ lifts to a path from level 0 to level $2m$, so it is not trivial for $m \neq 0$. ■

Theorem 10.3

(a) An arc does not separate S^2. (b) The Jordan separation theorem: a simple closed curve C separates S^2.

Idea. (a) If an arc D separated a from b, split D into two halves; by induction (bisecting the half that still separates), we get nested subarcs shrinking to a point whose complement-inclusions π_1 would all be nontrivial, contradicting that S^2 minus a point is simply connected. The key tool is: if $X = U \cup V$ with $U \cap V$ having exactly two path components, then $\pi_1(X)$ is nontrivial (a loop crossing between the components cannot be contracted), applied in the contrapositive. (b) Write $C = A_1 \cup A_2$ as two arcs meeting at two points. If $S^2 - C$ were connected, apply the same principle to $U = S^2 - A_1$, $V = S^2 - A_2$: $U \cup V = S^2 - \{p, q\}$ has infinite cyclic π_1, while by (a) and the nulhomotopy lemma the relevant loop would be trivial — a contradiction. ■

More generally, if D_1, D_2 are closed connected subsets of S^2 meeting in exactly two points and neither separates S^2, then $D_1 \cup D_2$ does.

10.2 Invariance of domain

Theorem 10.4 (Invariance of domain, $n = 2$)

If $U \subset \mathbb{R}^2$ is open and $f : U \to \mathbb{R}^2$ is continuous and injective, then $f(U)$ is open and f is an imbedding.

Idea. For a small closed disc B in U with boundary circle S, $f(S)$ is a simple closed curve and $f(B)$ is an arc-like non-separating set; $f(\operatorname{Int} B)$ is a connected set disjoint from $f(S)$ and must be one of the components of $\mathbb{R}^2 - f(S)$ (using the separation theorem and a π_1 argument showing the bounded component is filled), hence open. ■

Consequently $\mathbb{R}^2$ and $\mathbb{R}^n$ ($n \neq 2$) are not homeomorphic, and a 2-manifold cannot be an n-manifold for $n \neq 2$.

10.3 The Jordan curve theorem

Theorem 10.5 (Jordan curve theorem)

If C is a simple closed curve in S^2, then $S^2 - C$ has exactly two components, and C is the common boundary of each.

Idea. At least two components: Theorem 10.3. At most two: if U_1, U_2, U_3 were three components, using arcs and the fact that $\pi_1(S^2 - \{p, q\}) \cong \mathbb{Z}$ one produces a loop that is both nontrivial and trivial. Boundary: if C minus an open arc failed to be in the boundary of a component, the complementary arc would separate S^2, contradicting Theorem 10.3(a). ■

In $\mathbb{R}^2$ the two components are one bounded (the inside) and one unbounded. The **Schoenflies theorem** (not proved here) says the inside together with C is a closed disc. Both results fail in higher dimensions in the stronger form (Alexander horned sphere), though the separation statement persists.

10.4 Imbedding graphs in the plane

A (finite) **linear graph** is a space that is the union of finitely many arcs meeting at most at endpoints. A **theta space** is the union of three arcs with common endpoints.

Lemma 10.6

If $X \subset S^2$ is a theta space, $S^2 - X$ has exactly three components, with boundaries the three simple closed curves in X.

Theorem 10.7

The **utilities graph** $K_{3,3}$ (three houses joined to three utilities) cannot be imbedded in the plane. Neither can the complete graph K_5.

Idea. $K_{3,3}$ contains a theta space whose complement has three faces; the remaining vertex lies in one face whose boundary misses one of the three vertices it must be joined to, and the joining arc would have to cross the theta space. The argument for K_5 is similar. ■

(Kuratowski's theorem says these are the only obstructions.) Euler's formula $V - E + F = 2$ for connected planar graphs gives an alternative argument.

10.5 The winding number

For a loop $f : S^1 \to \mathbb{R}^2 - \{a\}$, the **winding number** $n(f, a)$ is the integer k such that $x \mapsto \frac{f(x)-a}{\|f(x)-a\|}$ is homotopic to $z \mapsto z^k$; equivalently the degree of this map, computed by lifting to $\mathbb{R}$.

Theorem 10.8

$n(f, a)$ is constant as a varies in a component of $\mathbb{R}^2 - f(S^1)$, and is 0 on the unbounded component. If f is a simple closed curve, then $n(f, a) = \pm 1$ for a inside and 0 outside.

Idea. Continuity in a and integrality give constancy on components; for a far away the normalized map has small image. For a simple closed curve, deform C near a point to a small circle around a and use the Jordan curve theorem to localize. ■

10.6 The Cauchy integral formula

For a piecewise smooth loop f in $\mathbb{C} - \{a\}$, $n(f, a) = \frac{1}{2\pi i} \int_f \frac{dz}{z-a}$: the integral computes the change of a continuous branch of $\log(z - a)$ along f, which is $2\pi i$ times the change of the lifted argument. Combined with Cauchy's theorem for simply connected domains (a holomorphic g on a simply connected region has $\int_f g = 0$ for every closed curve, by homotopy invariance of path integrals), this gives the Cauchy integral formula

$$n(f, a)\, g(a) = \frac{1}{2\pi i} \int_f \frac{g(z)}{z - a} dz,$$

and for a simple closed curve oriented counterclockwise $n(f, a) = 1$ at interior points a.

Chapter 11. The Seifert–van Kampen Theorem

The Seifert–van Kampen theorem computes the fundamental group of a space built from pieces whose groups are known. To state it we need some group theory.

11.1 Direct sums, free products and free groups

For abelian groups G_α, the **direct sum** $\bigoplus G_\alpha$ is characterized by the extension property: any family of homomorphisms $G_\alpha \to H$ (H abelian) extends uniquely to $\bigoplus G_\alpha \to H$. A **free abelian group** on $\{a_\alpha\}$ is $\bigoplus \mathbb{Z}a_\alpha$; its rank is well defined (rank of $G/2G$ as a $\mathbb{Z}/2$-vector space).

A group G generated by subgroups G_α with $G_\alpha \cap G_\beta = \{1\}$ is the **free product** $*G_\alpha$ if every element has a unique representation as a *reduced word* $g_1 \cdots g_n$, $g_i \neq 1$, consecutive letters from different G_α. Free products exist for any family (construct reduced words and let them act on the set of reduced words), and are characterized by the extension property: homomorphisms $G_\alpha \to H$ extend uniquely to $*G_\alpha \to H$ for every group H.

Theorem 11.1

If $G = G_1 * G_2$ and N_i is normal in G_i, then $G/\langle\langle N_1, N_2\rangle\rangle \cong (G_1/N_1) * (G_2/N_2)$, where $\langle\langle\cdot\rangle\rangle$ denotes the least normal subgroup containing the sets.

The **free group** on $\{a_\alpha\}$ is $*\langle a_\alpha\rangle$ with each $\langle a_\alpha\rangle \cong \mathbb{Z}$. Its abelianization $G/[G,G]$ is free abelian of the same rank, so free groups of different finite ranks are not isomorphic. A **presentation** $\langle a_\alpha \mid r_\beta\rangle$ is the quotient of the free group by the normal subgroup generated by the relators r_β.

11.2 The Seifert–van Kampen theorem

Theorem 11.2 (Seifert–van Kampen)

Let $X = U \cup V$ with U, V open and U, V, $U \cap V$ path connected, $x_0 \in U \cap V$. Let $\phi : \pi_1(U) * \pi_1(V) \to \pi_1(X)$ be induced by the inclusions. Then ϕ is surjective, and its kernel is the least normal subgroup containing all elements $i_1(g)^{-1} i_2(g)$, $g \in \pi_1(U \cap V, x_0)$, where i_1, i_2 are induced by $U \cap V \hookrightarrow U, V$. Equivalently, $\pi_1(X)$ is the pushout: homomorphisms $\pi_1(U) \to H$, $\pi_1(V) \to H$ agreeing on $\pi_1(U \cap V)$ combine uniquely to $\pi_1(X) \to H$.

Idea. Surjectivity is Theorem 9.11. For the kernel, given compatible ϕ_1, ϕ_2 define Φ on a loop by subdividing it into pieces lying in U or V, joining the break points to x_0, and multiplying the images. Independence of choices and of homotopy follows by subdividing a path homotopy $I \times I \to X$ into small squares lying in U or V and changing the factorization one square at a time. ∎

Corollary 11.3

If $U \cap V$ is simply connected, $\pi_1(X) \cong \pi_1(U) * \pi_1(V)$. If V is simply connected, $\pi_1(X) \cong \pi_1(U)/\langle\langle i_1 \pi_1(U \cap V)\rangle\rangle$.

11.3 The fundamental group of a wedge of circles

A **wedge of circles** is a Hausdorff space that is a union of subspaces $S_\alpha \cong S^1$ any two of which meet only in a common point p (with the coherent topology if there are infinitely many).

Theorem 11.4

The fundamental group of a wedge of circles S_α is free, with free generators represented by loops going once around each S_α.

Proof. For finitely many circles, induct with Corollary 11.3: thicken S_1 and $S_2 \vee \cdots \vee S_n$ by short arcs at p; their intersection is contractible. For infinitely many, use that a loop and a homotopy (compact) meet only finitely many circles, together with retractions onto finite subwedges. ■

So the figure eight has $\pi_1 \cong \mathbb{Z} * \mathbb{Z}$, which is non-abelian, and $\mathbb{R}^2$ minus n points (which deformation retracts to a wedge of n circles) has free fundamental group of rank n.

11.4 Adjoining a two-cell

Let $X = A \cup B$ with B a closed 2-cell attached along its boundary by $h : \partial B \to A$ (so X is A with a disc glued along the loop $f = h|_{S^1}$).

Theorem 11.5

The inclusion $A \hookrightarrow X$ induces a surjection $\pi_1(A, a) \to \pi_1(X, a)$ whose kernel is the least normal subgroup generated by the class of the attaching loop f (transported to a).

Proof. Apply Corollary 11.3 with V the open disc, $U = X$ minus the centre of the disc (which deformation retracts onto A), and $U \cap V$ an annulus with $\pi_1 \cong \mathbb{Z}$ generated by a loop parallel to f. ■

Attaching n-cells for $n \geq 3$ does not change π_1 (S^{n-1} is simply connected). Hence every group with a presentation is the fundamental group of a 2-dimensional complex: take a wedge of circles for the generators and attach a 2-cell for each relator.

11.5 The torus and the dunce cap

Example 11.1

(a) The torus is a wedge of two circles a, b with a disc attached along $aba^{-1}b^{-1}$: $\pi_1(T) \cong \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z} \times \mathbb{Z}$.
(b) The projective plane is a circle with a disc attached along a^2: $\pi_1(P^2) \cong \mathbb{Z}/2$, agreeing with Example 9.2.
(c) The n-fold **dunce cap** (a disc with its boundary wrapped n times around a circle) has $\pi_1 \cong \mathbb{Z}/n$. Every finite cyclic group, and indeed every finitely presented group, occurs as a fundamental group of a compact space.

Chapter 12. Classification of Surfaces

A **surface** is a 2-manifold. We show that every compact connected surface is homeomorphic to exactly one of: the sphere S^2, a connected sum of n tori T_n, or a connected sum of m projective planes P_m.

12.1 Fundamental groups of surfaces

A **labelling scheme** for a polygonal region (with an even number of edges) assigns to each edge a label and an orientation; identifying edges with the same label according to orientation gives a quotient space, which is a compact surface when each label occurs exactly twice. Schemes are written as words such as $aba^{-1}b^{-1}$.

Example 12.1

aa^{-1} gives S^2; $aba^{-1}b^{-1}$ the torus; aa the projective plane; $abab^{-1}$ the Klein bottle. The n-fold torus T_n comes from $a_1b_1a_1^{-1}b_1^{-1}\cdots a_nb_na_n^{-1}b_n^{-1}$, and the m-fold projective plane P_m from $a_1a_1a_2a_2\cdots a_ma_m$.

Theorem 12.1

If X is obtained from a polygonal region whose scheme identifies all vertices to one point, then X is a wedge of circles (one per label) with a 2-cell attached along the scheme word, so

$$\pi_1(T_n)\cong\langle a_1,b_1,\dots,a_n,b_n \mid [a_1,b_1]\cdots[a_n,b_n]\rangle,\qquad \pi_1(P_m)\cong\langle a_1,\dots,a_m \mid a_1^2\cdots a_m^2\rangle.$$

Proof. The boundary of the polygon maps onto the wedge of circles, the interior is an open 2-cell, and Theorem 11.5 applies. ∎

12.2 Homology of surfaces

The **first homology group** is the abelianization $H_1(X)=\pi_1(X)/[\pi_1(X),\pi_1(X)]$ (for path connected X; independent of the base point).

Theorem 12.2

$H_1(T_n)\cong\mathbb{Z}^{2n}$ and $H_1(P_m)\cong\mathbb{Z}^{m-1}\oplus\mathbb{Z}/2$. Consequently the surfaces $S^2,T_1,T_2,\dots,P_1,P_2,\dots$ are pairwise non-homeomorphic (indeed have non-isomorphic fundamental groups).

Proof. Abelianizing a presentation adds commutation relations. For T_n the relator becomes trivial, giving $\mathbb{Z}^{2n}$. For P_m the relation is $2(a_1+\cdots+a_m)=0$; changing basis to $a_1,\dots,a_{m-1},a_1+\cdots+a_m$ gives $\mathbb{Z}^{m-1}\oplus\mathbb{Z}/2$. ∎

12.3 Cutting and pasting

Two labelling schemes are **equivalent** if they give homeomorphic spaces. The following elementary operations preserve the homeomorphism type: (i) cutting a polygon along a diagonal into two polygons with a new label c on both sides, and its inverse (pasting); (ii) relabelling, and reversing the orientation of all occurrences of a label; (iii) cyclic permutation and inversion of the word; (iv) cancelling an adjacent pair aa^{-1} when the scheme has more than four letters, and its inverse.

Lemma 12.3

The **connected sum** $X\#Y$ (remove an open disc from each and glue along the boundary circles) has scheme w_1w_2 if X, Y have schemes w_1, w_2. Moreover $P_1\#T_1 \cong P_3$, i.e. $aabcb^{-1}c^{-1} \sim aabbcc$.

Idea. The second statement is a sequence of cuts and pastes that moves the pair aa past the commutator, turning $bcb^{-1}c^{-1}$ into two squares; it reflects that a Möbius band absorbs a handle into two cross-caps. ■

12.4 The classification theorem

Theorem 12.4 (Classification of surfaces, polygonal form)

Every space obtained from a polygonal region by a labelling scheme in which each label appears exactly twice is homeomorphic to S^2, T_n or P_m.

Idea. Using elementary operations: (1) cancel pairs aa^{-1} until either the scheme is aa^{-1} (sphere) or none remain; (2) reduce to a scheme in which all vertices are identified to one point (repeatedly cut off a triangle at a vertex of a class not equal to the most frequent one); (3) bring each *twisted pair* $\cdots a \cdots a \cdots$ together as aa; (4) any remaining *untwisted* pairs come in linked couples that can be grouped as commutators $aba^{-1}b^{-1}$; (5) if both aa and commutators appear, use $P_1\#T_1 \cong P_3$ to convert each commutator into two squares. The result is a normal form of type T_n or P_m. ■

Theorem 12.5

Every compact connected surface is homeomorphic to a quotient of a polygonal region as above (Radó: every compact surface can be triangulated), hence to exactly one of S^2, T_n $(n \geq 1)$, P_m $(m \geq 1)$.

The **Euler characteristic** $\chi = V - E + F$ of a triangulation is a topological invariant; $\chi(S^2) = 2$, $\chi(T_n) = 2 - 2n$, $\chi(P_m) = 2 - m$. Orientability distinguishes T_n from P_m, and χ then determines the surface. The Klein bottle is P_2.

12.5 Constructing compact surfaces

Compact surfaces with boundary are obtained by removing finitely many open discs; they are classified by orientability, Euler characteristic and the number of boundary circles. Every compact surface also arises from a polygon, from gluing triangles, or as a branched or unbranched cover of another surface; for instance a k-sheeted covering of T_n is $T_{k(n-1)+1}$ (the Euler characteristic is multiplied by k), and the orientation double cover of P_m is T_{m-1}.

Chapter 13. Classification of Covering Spaces

Throughout, $p : E \to B$ is a covering map with E and B path connected and locally path connected, $p(e_0) = b_0$. We classify such coverings by subgroups of $\pi_1(B, b_0)$.

13.1 Equivalence of covering spaces

Coverings $p : E \to B$ and $p' : E' \to B$ are **equivalent** if there is a homeomorphism $h : E \to E'$ with $p' \circ h = p$.

Lemma 13.1 (General lifting lemma)

Let Y be path connected and locally path connected and $f : (Y, y_0) \to (B, b_0)$ continuous. Then f lifts to $\tilde{f} : (Y, y_0) \to (E, e_0)$ iff $f_*\pi_1(Y, y_0) \subset p_*\pi_1(E, e_0)$; the lifting is unique.

Proof. Necessity is $f_* = p_*\tilde{f}_*$. For sufficiency define $\tilde{f}(y)$ as the endpoint of the lift (from e_0) of $f \circ \alpha$, where α is any path from y_0 to y; the subgroup condition makes it independent of α (the loop $f \circ (\alpha * \bar{\beta})$ lifts to a loop). Continuity uses local path connectedness of Y and evenly covered neighbourhoods. Uniqueness: the set where two liftings agree is open and closed. ∎

Theorem 13.2

p and p' are equivalent by an h with $h(e_0) = e_0'$ iff $H_0 = p_*\pi_1(E, e_0)$ and $H_0' = p_*'\pi_1(E', e_0')$ are equal. They are equivalent (with some base points) iff H_0 and H_0' are conjugate in $\pi_1(B, b_0)$.

Proof. Lift p through p' and p' through p (Lemma 13.1); the compositions are liftings of p, p' fixing base points, hence identities. Changing e_0 within the fibre along a path $\tilde{\alpha}$ conjugates H_0 by $[p \circ \tilde{\alpha}]$, and every conjugate arises. ∎

13.2 The universal covering space

A covering $q : Y \to B$ with Y simply connected is a **universal covering**. By Theorem 13.2 it is unique up to equivalence, and it covers every other covering: if $p : E \to B$ is a covering, there is a covering $r : Y \to E$ with $p \circ r = q$.

Example 13.1

$\mathbb{R} \to S^1$, $\mathbb{R}^2 \to T^2$, $S^2 \to P^2$, and $\mathbb{R}^2$ for every T_n $(n \geq 1)$ and P_m $(m \geq 2)$ (hyperbolic geometry for $n \geq 2$, $m \geq 3$). The universal covering of the figure eight is the infinite 4-valent tree.

13.3 Covering transformations

A **covering transformation** (deck transformation) is an equivalence $h : E \to E$ of p with itself; they form a group $\mathcal{C}(E, p, B)$. Let $H_0 = p_*\pi_1(E, e_0)$ and $N(H_0)$ its normalizer.

Theorem 13.3

$\mathcal{C}(E, p, B) \cong N(H_0)/H_0$. If H_0 is normal (the covering is **regular**), then $\mathcal{C}$ acts transitively on each fibre and $\mathcal{C} \cong \pi_1(B, b_0)/H_0$; in particular for the universal covering $\mathcal{C} \cong \pi_1(B, b_0)$.

Proof. For $e_1 \in p^{-1}(b_0)$ there is a deck transformation taking e_0 to e_1 iff $p_*\pi_1(E, e_1) = H_0$, i.e. iff the class $[\alpha]$ corresponding to e_1 under the lifting correspondence lies in $N(H_0)$; the map $[\alpha] \mapsto$ (the deck transformation sending e_0 to e_1) is a surjective homomorphism $N(H_0) \to \mathcal{C}$ with kernel H_0. ■

Conversely, if a group G acts on a path connected, locally path connected X **properly discontinuously** (every x has a neighbourhood U with $gU \cap U = \emptyset$ for $g \neq 1$), then $X \to X/G$ is a regular covering with deck group G, and $\pi_1(X/G)/p_*\pi_1(X) \cong G$. For simply connected X, $\pi_1(X/G) \cong G$: e.g. lens spaces $S^3/(\mathbb{Z}/n)$, and $\mathbb{R}^2/\mathbb{Z}^2 = T^2$.

13.4 Existence of covering spaces

B is **semilocally simply connected** if every point has a neighbourhood U such that loops in U are trivial in $\pi_1(B)$. This is necessary for a universal covering to exist.

Theorem 13.4

Let B be path connected, locally path connected and semilocally simply connected. For every subgroup $H \subset \pi_1(B, b_0)$ there is a covering $p : E \to B$ with $p_*\pi_1(E, e_0) = H$. Hence the equivalence classes of connected coverings correspond bijectively to conjugacy classes of subgroups of $\pi_1(B, b_0)$.

Construction. Let E be the set of equivalence classes of paths α in B starting at b_0, where $\alpha \sim \beta$ if they end at the same point and $[\alpha * \bar{\beta}] \in H$; let $p(\alpha^{\#}) = \alpha(1)$. Topologize E by the sets $B(U, \alpha) = \{(\alpha * \delta)^{\#} : \delta \text{ a path in } U \text{ from } \alpha(1)\}$, U a path connected neighbourhood. Semilocal simple connectivity makes p a homeomorphism on each $B(U, \alpha)$ for suitable U, and these sets are disjoint for inequivalent α over the same U, so p is a covering. The lift of a loop f ends at $f^{\#}$, which equals the base point iff $[f] \in H$; so $p_*\pi_1(E) = H$. ■

The **Hawaiian earring** (circles of radii $1/n$ tangent at a point) is not semilocally simply connected and has no universal covering.

Chapter 14. Applications to Group Theory

Covering space theory gives clean proofs of algebraic facts about free groups, by realizing free groups as fundamental groups of graphs.

14.1 Covering spaces of a graph

A **linear graph** is a space X that is a union of arcs (edges) meeting only at endpoints (vertices), with the coherent topology (a set is closed iff its intersection with each edge is closed). Graphs are Hausdorff, locally path connected and semilocally simply connected. A **tree** is a connected graph containing no simple closed curve (no cycle).

Theorem 14.1

A covering space of a graph is a graph: if $p : E \to X$ is a covering and X a graph, the lifts of edges and vertices make E a graph, and p maps edges homeomorphically onto edges.

Proof. Each edge is simply connected, so it lifts homeomorphically through every point of its preimage (general lifting lemma); the coherent topology on E follows from that on X and local homeomorphism. ■

14.2 The fundamental group of a graph

Lemma 14.2

Every tree is simply connected; indeed every finite tree, and any tree, deformation retracts onto a vertex (shrink one terminal edge at a time for finite trees; use the coherent topology in general).

Theorem 14.3

Every connected graph X has a maximal tree T (containing all vertices). If the edges not in T are $\{a_\alpha\}$, then $\pi_1(X)$ is free with one generator for each a_α. For a finite connected graph with V vertices and E edges, $\pi_1(X)$ is free of rank $1 - \chi(X) = E - V + 1$.

Proof. A maximal tree exists by Zorn's lemma (and contains every vertex, otherwise it could be extended). Collapsing T to a point is a homotopy equivalence (a tree is contractible and the pair has the homotopy extension property), and X/T is a wedge of circles, one for each a_α. Apply Theorem 11.4. A finite tree has $E_T = V - 1$ edges, so the number of extra edges is $E - V + 1$. ■

In particular the Euler characteristic $\chi(X) = V - E$ of a finite connected graph is determined by π_1, so it is a homotopy invariant.

14.3 Subgroups of free groups

Theorem 14.4 (Nielsen–Schreier)

Every subgroup H of a free group F is free.

Proof. Realize F as π_1 of a wedge of circles X (a graph). By Theorem 13.4 there is a covering $p : E \to X$ with $p_*\pi_1(E) = H$, and p_* is injective, so $H \cong \pi_1(E)$. By Theorem 14.1, E is a graph, and by Theorem 14.3 its fundamental group is free. ■

Theorem 14.5 (Schreier index formula)

If F is free of finite rank n and $H \subset F$ has finite index k, then H is free of rank $k(n-1)+1$.

Proof. The covering E is a k-sheeted cover of the wedge X of n circles (one vertex, n edges), so E has k vertices and kn edges, and $\operatorname{rank}\pi_1(E) = kn - k + 1$ by Theorem 14.3. ■

Example 14.1

(a) The commutator subgroup of $\mathbb{Z} * \mathbb{Z} = F(a,b)$ is free of infinite rank: it corresponds to the covering of the figure eight by the integer grid in $\mathbb{R}^2$ (the abelianization $\mathbb{Z}^2$ is the deck group), whose fundamental group is free on infinitely many generators.
(b) A free group of rank 2 contains free subgroups of every finite rank and of countably infinite rank; e.g. the kernel of $F(a,b) \to \mathbb{Z}/2$, $a, b \mapsto 1$, has index 2 and rank 3.
(c) In contrast, subgroups of $\mathbb{Z}^n$ are free abelian of rank at most n; for free groups the rank of a finite-index subgroup is larger than that of the group (when $n \geq 2$).